



KSR College of
Engineering

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AI-Based Gesture Control for Improving Navigation Accuracy in Telepresence Robotic System Using Deep Learning

Guide:

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Identification of Problem Domain

- **Assistive Robotic Control:** People with limited mobility or physical interaction capabilities may require intuitive and accessible methods to control robotic systems.
- **Gesture-Based Interaction:** Conventional robotic control methods such as buttons, joysticks and wired interfaces can be difficult to use for natural human–robot interaction.
- **Sensor Integration:** Flex sensors, IMU sensors and other inputs generate multiple real-time signals that must be processed and mapped accurately to robotic movements.
- **Real-Time Response:** Delays in sensor processing, communication and AI decision-making can affect the accuracy and usability of the robotic system.
- **Intelligent Interaction:** A robotic system should combine gesture control, voice-based AI, environmental sensing and autonomous responses for more natural interaction.
- **Research Need:** An effective framework should integrate **wearable sensing, wireless communication, robotic actuation and AI-based interaction** into a reliable real-time assistive robotic platform.

Objectives and Purpose of Project

- Develop a **wearable glove-based robotic control system**.
- Capture movements of all **five fingers** using flex sensors.
- Track hand and arm orientation using **IMU sensors**.
- Establish reliable **wireless communication using ESP32 and ESP-NOW**.
- Integrate voice-based **AI interaction, speech recognition and speech synthesis**.
- Develop a responsive robotic platform capable of performing user-controlled movements.

Quality & Relevance of Project Selection

- **Real-World Relevance:** CYNEXIS addresses the need for intuitive and accessible human–robot interaction.
- **Innovative Approach:** Combines **gesture control, voice AI, wireless communication and robotic actuation** in a single platform.
- **Technical Feasibility:** Uses readily available components such as **ESP32, flex sensors, MPU6050 and servo motors**.
- **Interdisciplinary Scope:** Integrates **embedded systems, robotics, artificial intelligence, IoT and sensor technology**.
- **Practical Application:** The system can support **assistive robotics, remote robotic control and natural user interaction**.
- **Future Potential:** The modular architecture allows additional sensors, AI capabilities and robotic functions to be integrated.

Literature Survey - Comparative Analysis

Author & Year	Methodology Used	Key Findings	Identified Limitations
Salman et al. (2020)	Flex-sensor glove + wireless robotic arm	Five flex sensors successfully controlled robotic finger movements wirelessly.	Limited sensing of hand orientation and mainly focused on robotic hand control.
Škulj et al. (2021)	Wearable IMU-based teleoperation	IMUs enabled real-time upper-body motion tracking for collaborative robot control.	Focused primarily on IMU-based motion and did not provide five-finger flex sensing.
Herbaz et al. (2025)	Flex sensors + MPU6050 + sensor fusion + RNN	Achieved real-time gesture classification using multimodal wearable sensing.	Requires machine-learning processing and focuses mainly on gesture classification.
Proposed Work — CYNEXIS	Five flex sensors + MPU6050 + ESP32/ESP-NOW + AI voice interaction.	Combines gesture sensing, wireless robotic control and AI-based voice interaction in one platform.	Requires complete sensor calibration and integration of the robotic and AI subsystems.

Literature Survey - Key Takeaways

- **Gesture-Based Control** – Flex sensors and IMU enable intuitive robotic control.
- **Wireless Communication** – ESP32 and ESP-NOW provide fast communication between glove and rover.
- **Voice-Based AI** – Speech recognition and local AI enable natural voice interaction.
- **Robotic Integration** – Combines gesture and voice commands for rover and robotic-arm control.
- **Real-Time Response** – Optimized AI pipeline reduces response latency.
- **System Gap** – Existing approaches often focus on either voice or gesture; CYNEXIS combines both into one integrated system.

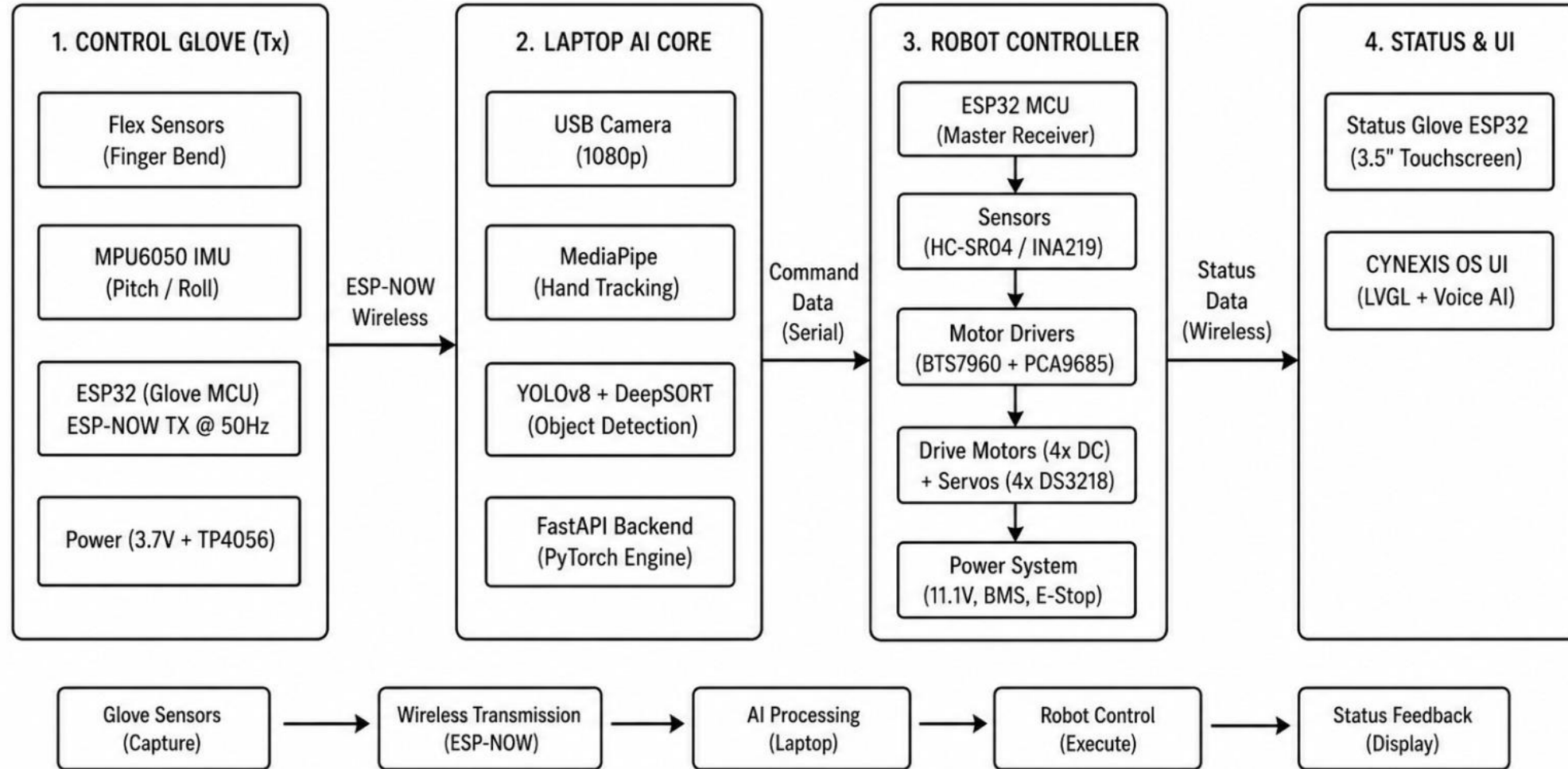
Study of Existing Systems

- **Voice-Controlled Robots** – Use speech recognition to control robotic functions.
- **Gesture-Controlled Robots** – Use flex sensors and IMU sensors to interpret human gestures.
- **Wireless Robotic Systems** – ESP32-based systems provide wireless communication and control.
- **AI-Based Assistants** – Local AI models provide intelligent voice interaction and responses.
- **Existing Limitation** – Most systems focus on a single interaction method rather than combining voice and gesture control.
- **CYNEXIS Approach** – Integrates **Voice AI + Gesture Control + Wireless Communication + Rover/Robotic Arm** into a single system.

Existing Systems vs. Proposed Solution

Evaluation Feature	Existing / Legacy Systems	Proposed Framework
Architecture Type	Monolithic & Rigid	Microservices & Modular Pipeline
Processing Overhead	High computational consumption	Optimized lightweight algorithmic execution
Adaptability	Manual rule updates required	Autonomous real-time learning adaptation
Fault Tolerance	Single Point of Failure (SPOF)	Self-healing redundant node failover

Methodology Overview



Implementation Timeline & Milestones

Phase	Milestone
Phase 1	Project Planning
Phase 2	AI Software Development
Phase 3	Control Glove Development
Phase 4	Rover & Robotic Arm Assembly
Phase 5	ESP32 & Sensor Integration
Phase 6	Voice + Gesture Integration
Phase 7	System Testing & Calibration
Phase 8	Final Demonstration

Questions & Discussion

THANK YOU